

# HENRY HSU

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## SUMMARY

AI engineer working on both sides of agent systems: shipping them, and attacking them. Reproduced a tool-schema poisoning attack on Llama 3.1-8B and built a representation-level defense that cut privacy leakage 14.5pp with no loss in benign task success. Previously GenAI Team Lead at MaiAgent, where production AI agents with tool APIs and MCP grew users from 3K to 20K and cut token usage 67%. 14 merged PRs to LlamaIndex. M.C.S. at Rice, graduating Dec. 2026, available Feb. 2027.

## WORK EXPERIENCE

### Software Engineering Intern

Jun. 2026 – Aug. 2026

Google

Taipei, Taiwan

- **Integration Test Infrastructure:** Built 3 test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage:** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction:** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

### Generative AI Team Lead

Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. – Award-Winning B2B GenAI Startup

Taipei, Taiwan

- **Building AI Agents:** Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source:** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework:** Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership:** Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

### Generative AI Intern

Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. – Award-Winning B2B GenAI Startup

Taipei, Taiwan

- **Pipeline Scaling:** Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

## EDUCATION

### Rice University - Top 20 U.S. University

Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00)

Houston, TX

### National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University

Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List)

Hsinchu, Taiwan

Minor: Computer Science (Domain GPA: 4.13/4.30)

## PROJECTS

### Defending LLM Agents Against Tool Schema Poisoning | PyTorch, Llama 3.1, Activation Steering, Slurm

Apr. 2026 – May 2026

- **Threat Model:** Reproduced a tool-schema poisoning attack where deceptive parameter descriptions make structured-output agents auto-fill private data, setting a **56%** attack-success and **52%** privacy-leak baseline on Llama 3.1-8B
- **Representation-Level Defense:** Probed hidden states to locate a linear safety direction (**96%** probe accuracy at layer 10) and steered activations along it at inference, cutting privacy leakage **14.5pp** with no loss in benign task success
- **Defense Comparison:** Ran a 2×2 single/multi-turn factorial showing the in-context reflection baseline buys its leak reduction with a **6pp** higher attack rate and **6pp** lower task success, both of which steering avoids

## AWARDS

**Presidential Hackathon Winner (2024, Top 5 Nationally)** - Urban noise agent: structured-output LLM pipeline and tool calling

**23rd Golden Peak Award (2025)** - Outstanding Commercial Product, MaiAgent AI Platform

## SKILLS

**Languages:** Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Backend:** Django, FastAPI

**AI Security:** Activation Steering, Hidden-State Probing, Red-Team Evaluators, Structured-Output Attack Surfaces, Tool-Call Guardrails

**ML:** PyTorch, numpy, pandas, scikit-learn | **Data:** PostgreSQL, Milvus, Elasticsearch | **Infra:** Docker, git, Slurm, AWS Bedrock/EC2/S3